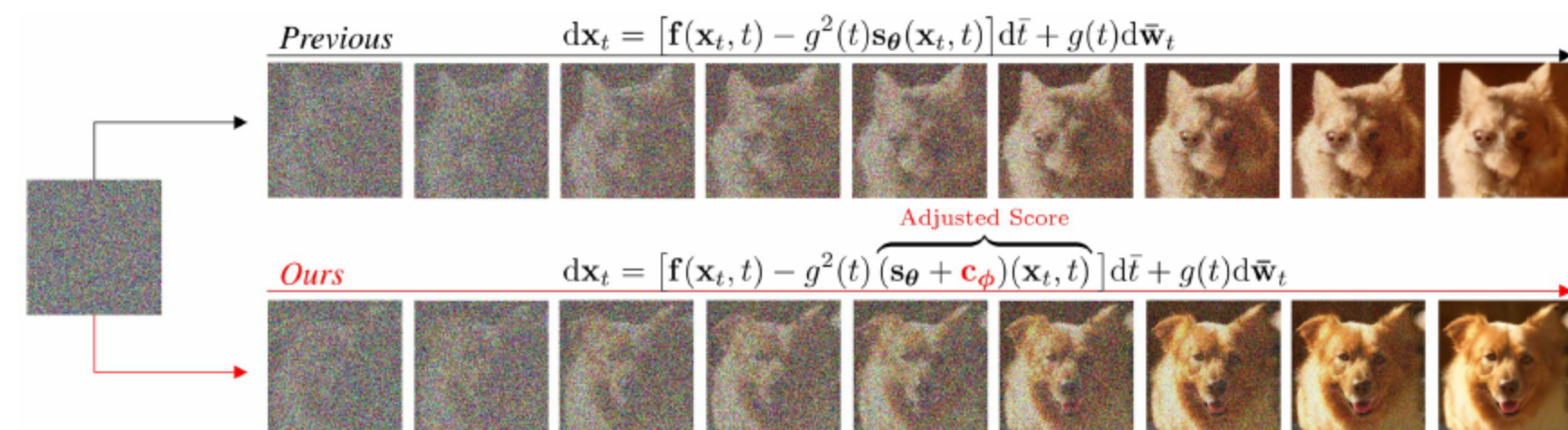


Refining Generative Process with Discriminator Guidance in Score-based Diffusion Models

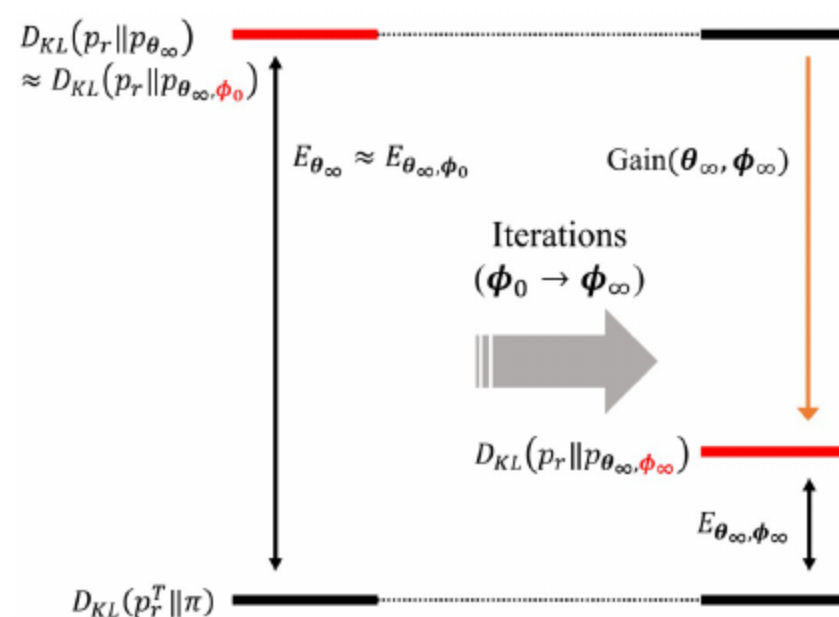
Dongjun Kim; Yeongmin Kim; Se Jung Kwon;
Wanmo Kang; Il-Chul Moon

01 Guidance



- Pretrained scores are reused widely across image, video, and text-to-image tasks; further score training risks **overfitting and memorization**
- Training fresh scores is costly; we need a post-hoc route to boost **realism and diversity** without touching the base model
- **Goal:** refine samples from a fixed score model using a cheap, stable auxiliary signal learned once
- Downstream users benefit from **plug-in guidance** that preserves the original model while enhancing quality

02 Fixed Score + Discriminator Guidance

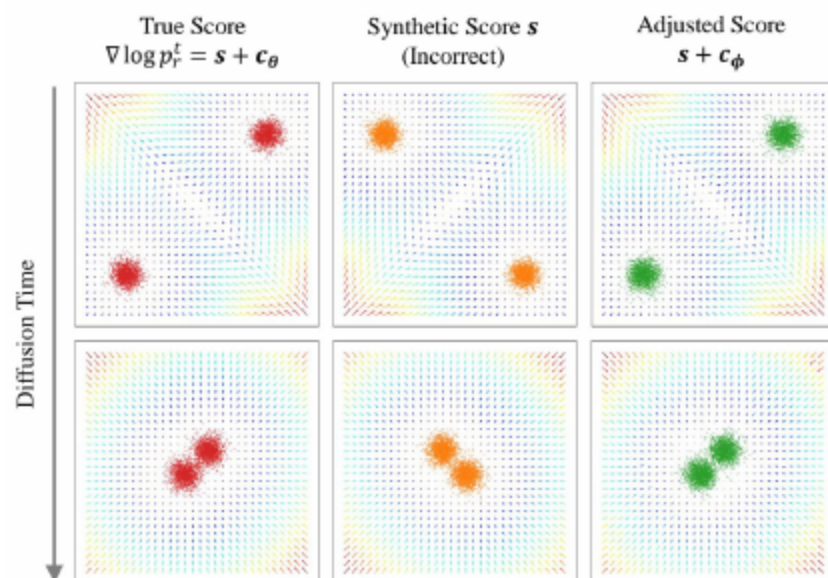


- Train a time-conditioned discriminator to estimate a likelihood ratio across noise scales over real versus generated paths
- Add a correction term to the fixed score during sampling to steer trajectories toward realistic regions
- At an optimal discriminator, the adjusted score aligns with the data score, yielding **more faithful** denoising
- **Key property:** guidance is applied only at sampling time, leaving the base score network unchanged

03 Contributions and Teaser Visual

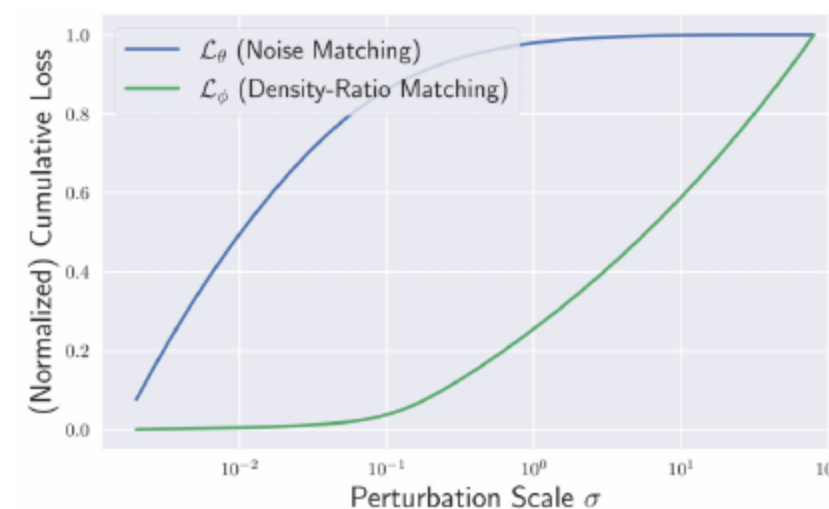


- Discriminator guidance improves both **fidelity and diversity** and complements classifier guidance
- New state of the art on CIFAR-10, CelebA or FFHQ, and ImageNet across multiple metrics
- Stable training of a small discriminator brings fast convergence with a fixed score
- **Practical plug-in:** minimal changes to base pipelines in data or latent diffusion



- Forward diffusion stochastically perturbs data; the reverse process uses a learned score to denoise
- Generation follows the reverse-time dynamics driven by the score network
- Score errors distort reverse trajectories, harming **realism and coverage**
- **Motivation:** correct residual score bias post-hoc without retraining the full model

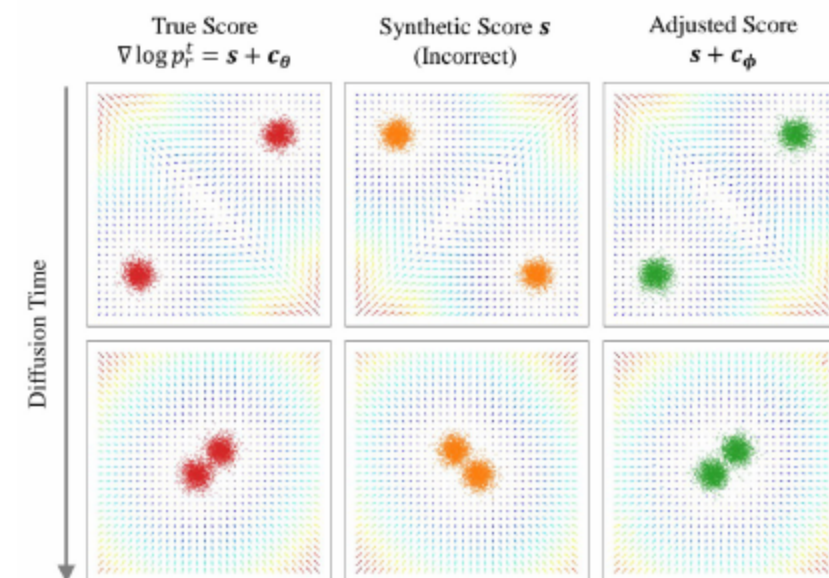
[Song et al., 2020b; Anderson, 1982; Ho et al., 2020]



- Denoising score matching targets the data score of the perturbed distribution at each time
- With a standard temporal weight, the loss equals a joint Kullback–Leibler divergence over full trajectories
- Residual errors are typically larger at high noise scales, motivating post-hoc refinement
- **Contrast:** our method refines a fixed score via noise contrastive estimation rather than retraining

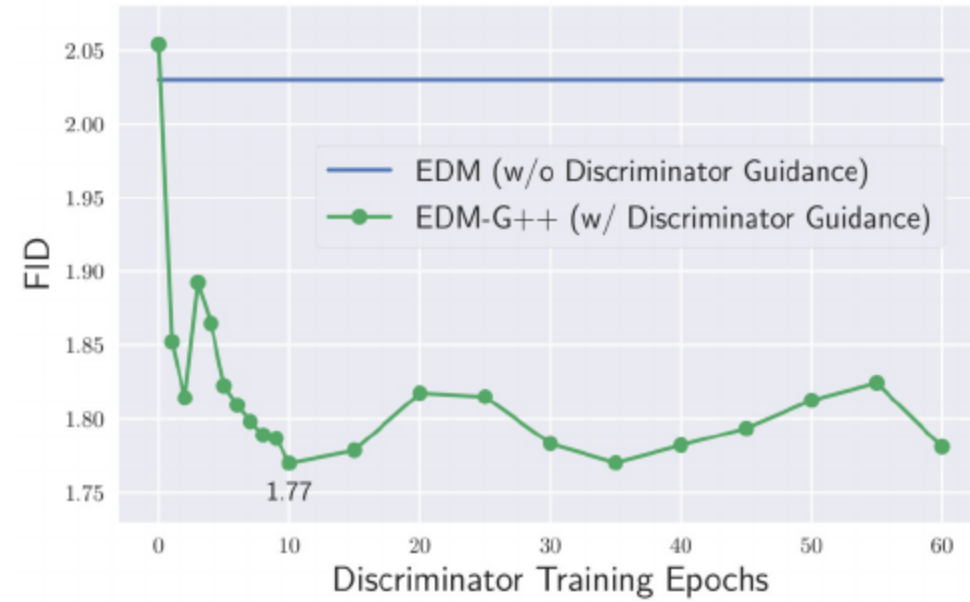
[Song & Ermon, 2019; Kingma et al., 2021; Kim et al., 2022b]

After training, generation follows the reverse dynamics with a fixed score. We refine sampling by augmenting the fixed score with a learned correction, aiming to align the model's reverse process with the data reverse process while avoiding any further score training.



- Under mild conditions, the reverse process with an adjusted score can match the data reverse process
- The ideal correction equals the gradient of the likelihood ratio between data and model marginals at each time
- Direct likelihood ratios are intractable; we estimate them with a trained discriminator
- **Implication:** perfect ratio recovery removes score error, yielding **exact** reverse dynamics

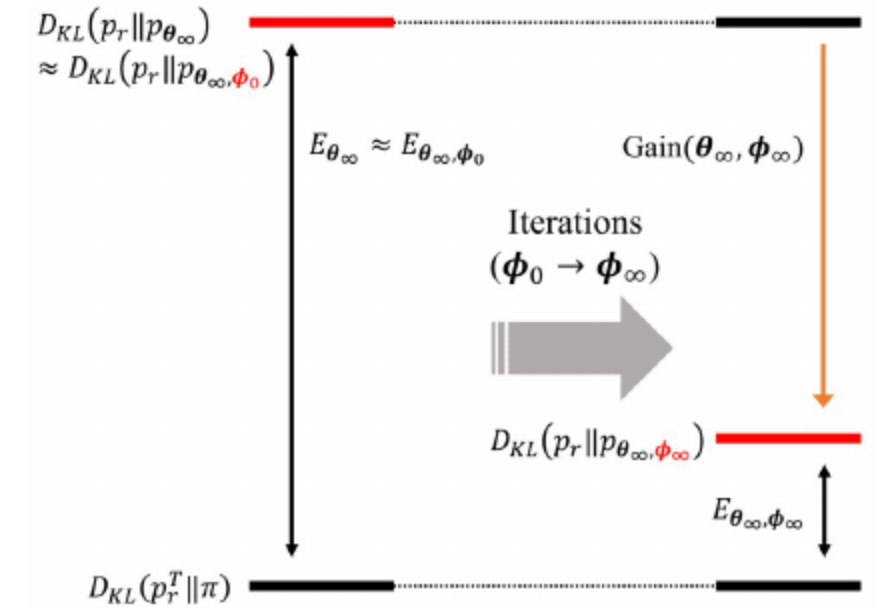
08 Discriminator Training and DG Definition



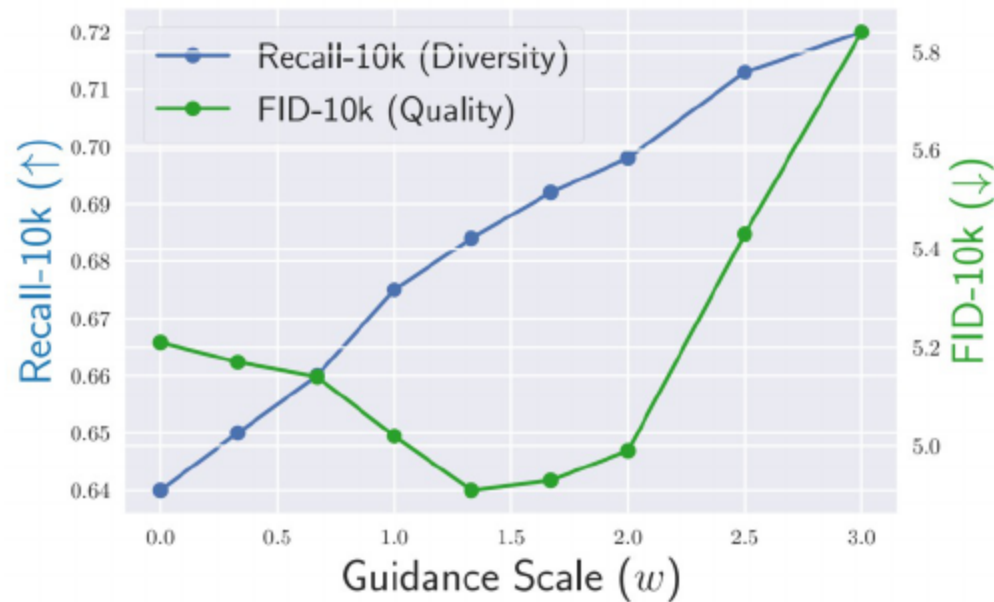
- Train a time-embedded discriminator with binary cross entropy to classify real forward paths versus generated reverse paths
- Convert the discriminator output into a likelihood ratio estimate and take its gradient
- Define discriminator guidance by adding a weighted ratio-gradient to the model score during sampling
- Control knob:** a time-dependent weight balances quality and diversity

09 Theory and Gain

- Discriminator guidance reduces divergence to data by shrinking the adjusted score error
- The gain grows as the discriminator improves, starting near zero when untrained
- At an optimal discriminator, adjusted score error vanishes, maximizing **closeness** to data
- Takeaway:** learning the ratio field directly translates into better reverse trajectories



10 Convergence, Control, and CG Context



- Discriminator training is a stable minimization with quick convergence since the score network is frozen
- A time-dependent weight trades off **fidelity** and **diversity**, with a sweet spot for balanced metrics
- Combining with classifier guidance restores diversity while retaining strong fidelity
- Outcome:** improved precision-recall balance versus classifier-only baselines

[Dhariwal & Nichol, 2021]

11 Experimental Setup and Discriminator Architecture

	Cifar-10			Cifar-100		ImageNet	
	LSGM-G++	EDM-G++	EDM	Soft Truncation-G++	EDM-G++	ADM-G++	DT-G++
Pre-trained Score Network							
Model	LSGM	EDM	EDM	Soft Truncation	EDM	ADM	DT
Class condition	✓	✓	✓	✓	✓	✓	✓
Discriminator Training							
Model	UVP	UVP	UVP	CVP	CVP	UVP	UVP
Class condition	✓	✓	✓	✓	✓	✓	✓
Time sampling	Importance	Importance	Importance	Importance	Importance	Importance	Importance
λ	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}
Maximum diffusion time	128	128	128	128	128	128	128
EMA	✓	✓	✓	✓	✓	✓	✓
Batch size	128	128	128	128	128	128	128
# of	50,000	50,000	50,000	50,000	50,000	50,000	50,000
# of	50,000	50,000	50,000	50,000	50,000	50,000	50,000
# of Epochs	200	200	200	200	200	200	200
GPUs	1x V100	1x V100	1x V100	1x V100	1x V100	1x A100	1x A100
Pre-trained Classifier							
Model	No classifier	ADM	ADM	ADM	ADM	ADM	ADM
Input shape (data dimension)		(B,3,32,32)	(B,3,32,32)	(B,3,32,32)	(B,3,32,32)	(B,3,32,32)	(B,3,32,32)
Output shape (latent dimension)		(B,3,32,32)	(B,3,32,32)	(B,3,32,32)	(B,3,32,32)	(B,3,32,32)	(B,3,32,32)
Shallow U-Net Architecture							
Input shape (latent dimension)	(B,16,160)	(B,16,160)	(B,16,160)	(B,16,160)	(B,16,160)	(B,16,160)	(B,16,160)
Output shape	(B,1)	(B,1)	(B,1)	(B,1)	(B,1)	(B,1)	(B,1)
Minimum value of discriminator (log)	10^{-5}	0	0	0	0	0	0
Maximum value of discriminator (log)	1×10^{-5}	1	1	1	1	1	1
Class condition	✓	✓	✓	✓	✓	✓	✓
# of Epochs	5	4	4	5	5	4	4
# of Attention Heads	5	5	5	5	5	5	5
Activation resolution	16, 8	8	8	8	8	8	8
Model channel	128	128	128	128	128	128	128
Channel multiplier	1	1	1	1	1	1	1
Sampling							
Model	UVP	WVE	WVE	UVP	WVE	UVP	UVP
Class condition	✓	✓	✓	✓	✓	✓	✓
Minimum value of discriminator (log)	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}
Maximum value of discriminator (log)	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
Subnet	PRODE	PRODE	PRODE	PRODE	PRODE	PRODE	PRODE
Subnet accuracy of μ_p	1 st -order	2 nd -order	2 nd -order	1 st -order	2 nd -order	2 nd -order	2 nd -order
Subnet type of μ_p	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU
Subnet accuracy of σ_p	1 st -order	1 st -order	1 st -order	1 st -order	1 st -order	1 st -order	1 st -order
Subnet type of σ_p	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU
$\mu_{p,1}$	0.1	0.0	0.0	0.0	0.0	0.1	0.1
$\sigma_{p,1}$	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Classifier Guidance	✓	✓	✓	✓	✓	✓	✓
EDM	✓	✓	✓	✓	✓	✓	✓
$\mu_{p,1}^{(2)}$	0	0	0	0	0	0	0
$\sigma_{p,1}^{(2)}$	2	2	Adaptive	Adaptive	Adaptive	Adaptive	Adaptive

- Use a time-embedded U-Net encoder as discriminator; freeze pretrained classifier features and fine-tune a shallow encoder
- Support both data and latent diffusion: train from scratch when required, otherwise fine-tune for efficiency
- Class-conditional discriminator for conditional datasets; lightweight to deploy
- Choice rationale:** U-Net encoder yielded the best guidance among tested backbones

[Dhariwal & Nichol, 2021; Peebles & Xie, 2022; Vahdat et al., 2021]
[Karras et al., 2022]

Model	Diffusion Space	NFE↓	Unconditional NLL↓	FID↓	Conditional FID↓
VDM (Kingma et al., 2021)	Data	1000	2.49	7.41	-
DDPM (Ho et al., 2020)	Data	1000	3.75	3.17	-
iDDPM (Nichol & Dhariwal, 2021)	Data	1000	3.37	2.90	-
Soft Truncation (Kim et al., 2022b)	Data	2000	2.91	2.47	-
INDM (Kim et al., 2022a)	Latent	2000	3.09	2.28	-
CLD-SGM (Dockhorn et al., 2022)	Data	312	3.31	2.25	-
NCSN++ (Song et al., 2020b)	Data	2000	3.45	2.20	-
LSGM (Vahdat et al., 2021)	Latent	138	3.43	2.10	-
NCSN++-G (Chao et al., 2022)	Data	2000	-	-	2.25
EDM [†] (random seed)	Data	39	2.60	2.03	1.82
EDM (reported, manual seed)	Data	35	2.60	1.97	1.79
LSGM-G++	Latent	138	3.42	1.94	-
EDM-G++	Data	35	2.55	1.77	1.64

- State-of-the-art FID on CIFAR-10, CelebA or FFHQ, and ImageNet with discriminator guidance
- Gains hold in both data diffusion and latent diffusion settings with the same base hyperparameters
- Improvements are attributable to guidance rather than heavy retraining
- **Robustness:** consistent benefits across model families and resolutions

Model	Diffusion Space	FID↓	sFID↓	IS↑	Prec↑	Rec↑	F1↑
Validation Data		1.68	3.67	232.21	0.75	0.66	0.70
ADM (Dhariwal & Nichol, 2021)	Data	10.94	6.02	100.98	0.69	0.63	0.66
DiT-XL/2 (Peebles & Xie, 2022)	Latent	9.62	6.85	121.50	0.67	0.67	0.67
ADM-G (Dhariwal & Nichol, 2021)	Data	4.59	5.25	186.70	0.82	0.52	0.64
RIN (Jabri et al., 2022)	Data	4.51	-	161.00	-	-	-
LDM-4-G (Rombach et al., 2022)	Latent	3.60	-	247.67	0.87	0.48	0.62
RIN + schedule (Chen, 2023)	Data	3.52	-	186.20	-	-	-
simple diffusion (Hooeboom et al., 2023)	Data	2.77	-	211.80	-	-	-
DiT-XL/2-G (Peebles & Xie, 2022)	Latent	2.27	4.60	278.24	0.83	0.57	0.68
ADM-G++	Data	3.18	4.53	255.74	0.84	0.53	0.66
DiT-XL/2-G++	Latent	1.83	5.16	281.53	0.78	0.64	0.70

- Best FID and recall with a large latent transformer plus guidance; strong sFID and inception score as well
- Balanced precision-recall, with recall notably improved over classifier-only baselines
- Validation data references contextualize ceilings in inception score and precision
- **Net effect:** higher quality and **broader coverage** on ImageNet classes

[Hooeboom et al., 2023; Jabri et al., 2022; Chen, 2023]

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- Guidance adds light overhead versus score training; discriminator converges quickly
- Extra cost is additional sampling plus discriminator updates with modest GPU hours
- Component-wise budgets indicate **low carbon and compute** relative to training a new score
- **Deployment:** plug-in discriminator can be trained offline and reused

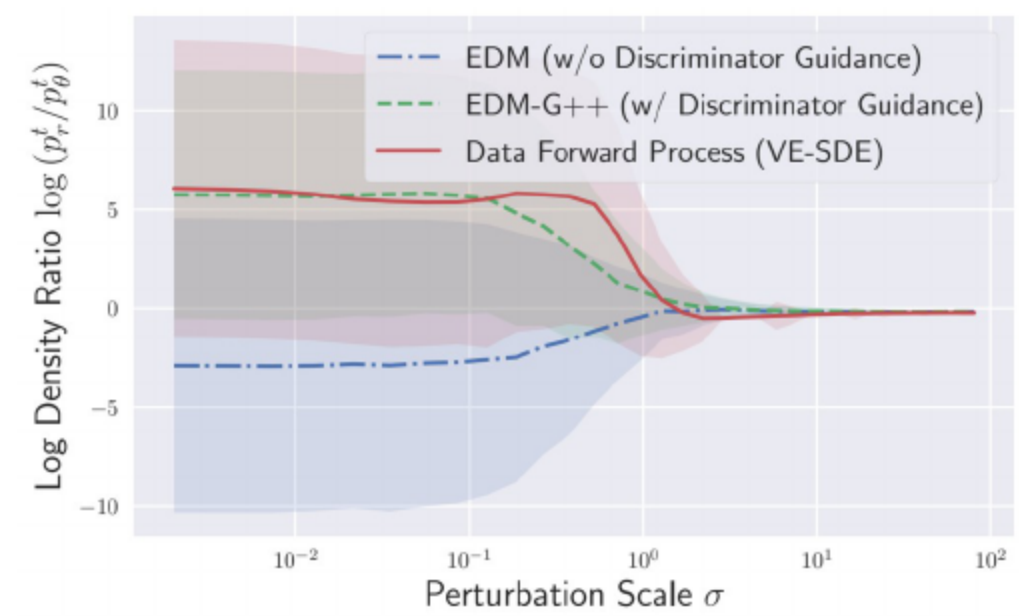
- Average reconstructions move closer to originals under guidance for small perturbations
- Expectation analysis via Tweedie's formula indicates improved score accuracy with guidance
- Face dataset regeneration FID improves substantially with the adjusted score
- **Signal:** guidance reduces bias around local neighborhoods in image space



[Robbins, 1992; Meng et al., 2021]

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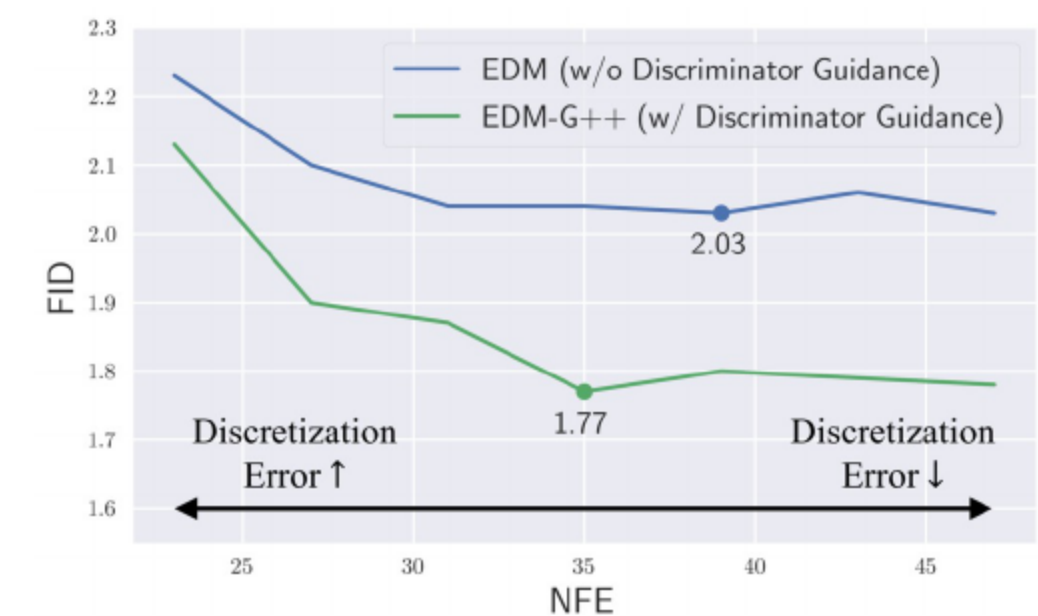
Trajectory Likelihood-Ratio Alignment



- Adjusted denoising paths align with the likelihood-ratio profile of real forward diffusion
- The trained discriminator cannot distinguish adjusted paths from real trajectories
- Supports correctness of the estimated correction from the ratio gradient
- Interpretation:** trajectory-level alignment explains downstream metric gains

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NFE Sensitivity and Training Signal



- Gains diminish at very low function evaluations where discretization error dominates **sampling quality**
- Discriminator loss emphasizes large diffusion times, capturing diversity-sensitive errors
- Future work includes adapting guidance to extremely small-step solvers and specific solver stages
- Practical tip:** apply stronger weights at early noisy stages and reduce at later clean stages

[De Bortoli, 2022]

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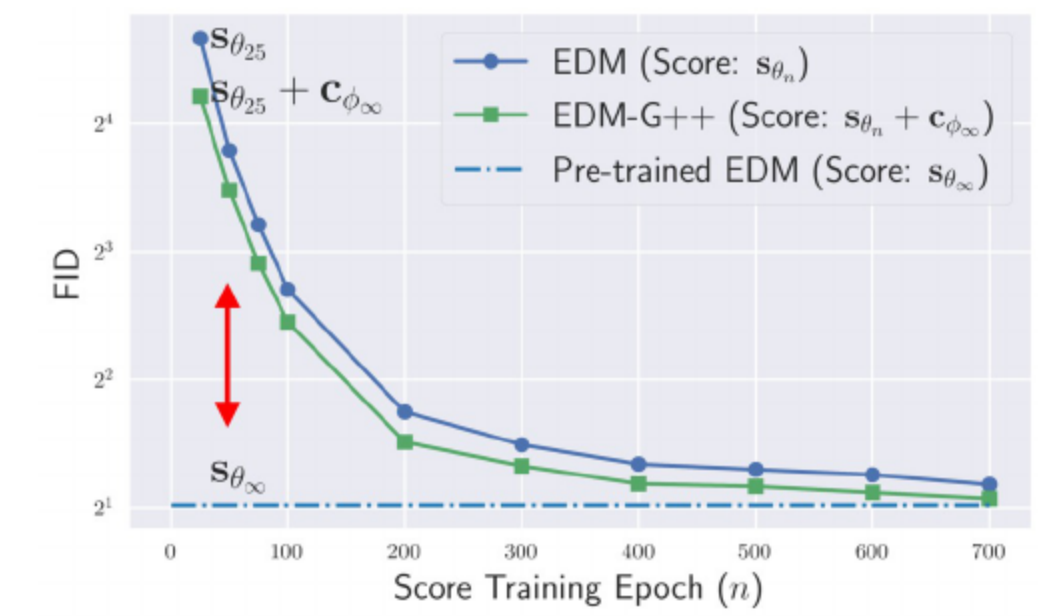
Key Takeaways and Divergence View

Model	Density-Ratio Matching $D_h(r_{\theta_\infty} \ r_\phi)$	FID
EDM	-	2.03
EDM-G++	UKL (Nguyen et al., 2010)	1.84
	$h(r) = r \log r - r$	
	LSIF (Kanamori et al., 2009)	1.84
	$h(r) = (r - 1)^2 / 2$	
	BCE ($\mathcal{L}_\phi$ of Eq. 4)	1.77
	$h(r) = r \log r - (r + 1) \log (1 + r)$	

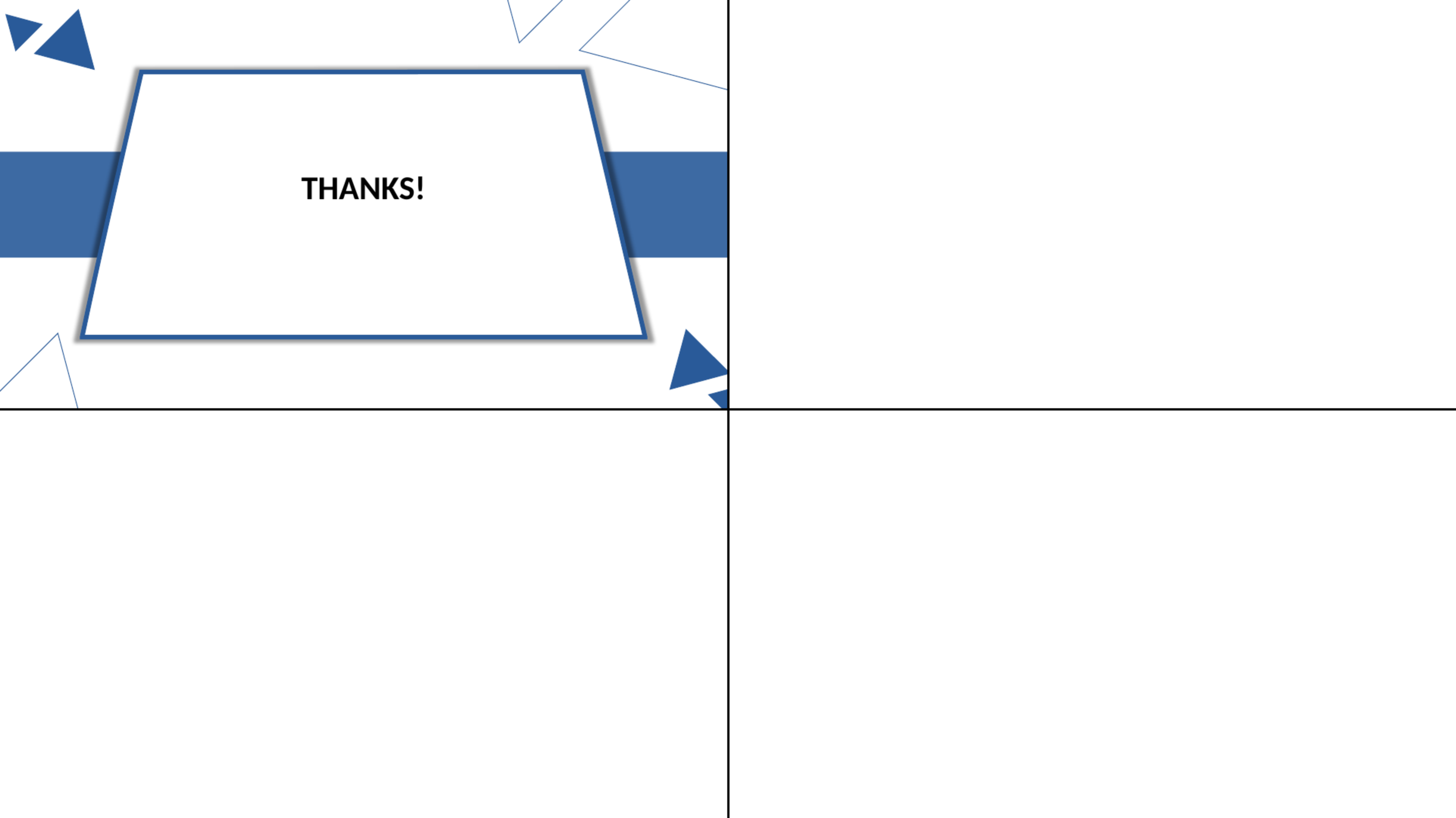
- Binary cross entropy belongs to both h-Bregman and f-divergence families; alternatives like least squares importance fitting and unnormalized Kullback-Leibler yield similar gains
- Viewing guidance as density-ratio matching suggests broader divergence-based refinements
- Weighting via f-divergences can prioritize perceptually plausible regions with high data-to-model ratio
- Design space:** choose divergences and weights to target desired precision-recall trade-offs

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Limitations and Future Work



- Simultaneous training of score and discriminator shows limited gains due to decoupled objectives
- Low function evaluation regimes remain challenging; guidance impact is reduced **in practice**
- Exploring joint objectives and alternative divergences may further improve practicality
- Open path:** solver-aware guidance schedules to mitigate discretization artifacts

The background features a white canvas with a thin black horizontal line across the middle and a thin black vertical line on the right side. In the upper-left quadrant, there are several blue geometric elements: a small triangle, a larger triangle, a horizontal rectangle, and a parallelogram. The word "THANKS!" is centered within this parallelogram. In the upper-right quadrant, there are thin blue lines forming a larger triangle and a smaller triangle. In the lower-right quadrant, there is a small blue triangle pointing towards the center.

THANKS!